Algorithms for Modern Processor Architectures

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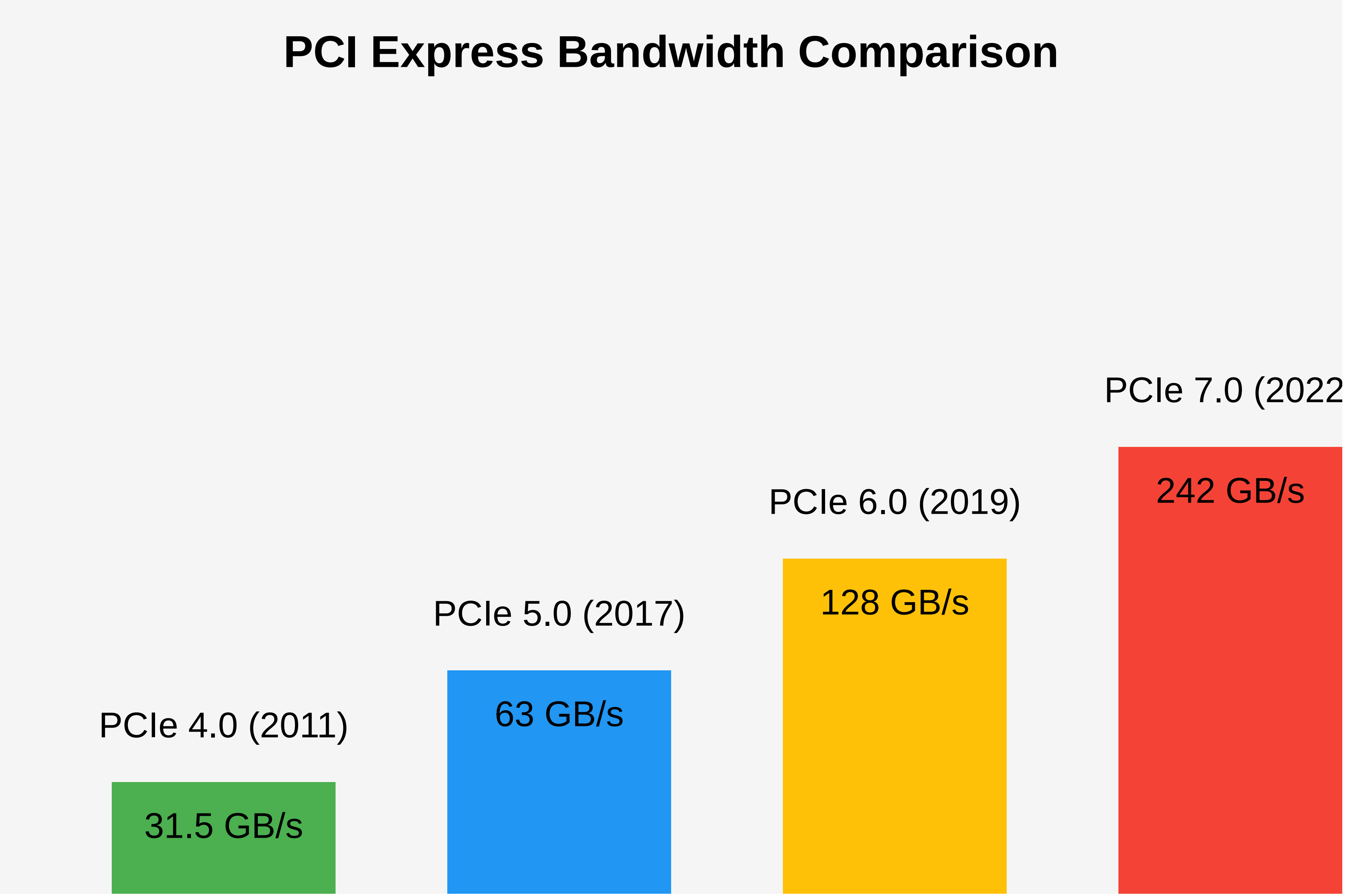
All software for this talk: https://github.com/lemire/talks/tree/master/2025/sea/software

Disk at gigabytes per second

Micron shows off world's fastest PCIe 6.0 SSD, hitting 27 GB/s speeds — Astera Labs PCIe 6.0 switch enables impressive sequential reads

News By Sunny Grimm published March 8, 2025

The next-gen of networking and storage is hitting the trade shows


PCI Express Bandwidth Comparison
PCIe 4.0 (2011)
31.5 GB/s
PCIe 5.0 (2017)
63 GB/s
PCIe 6.0 (2019)
128 GB/s
PCIe 7.0 (2022
242 GB/s

High Bandwidth Memory

- Xeon Max processors contain 64 GB of HBM
- Bandwidth 800 GB/s

Wikipedia	Wikipedia	Wikipedia	Wikipedia	Wikipedia	Wikipedia	Wikipedia	Wikipedia
Wikipedia	Wikipedia	Wikipedia	Wikipedia	Wikipedia	Wikipedia	Wikipedia	Wikipedia
Wikipedia	Wikipedia	Wikipedia	Wikipedia	Wikipedia	Wikipedia	Wikipedia	Wikipedia
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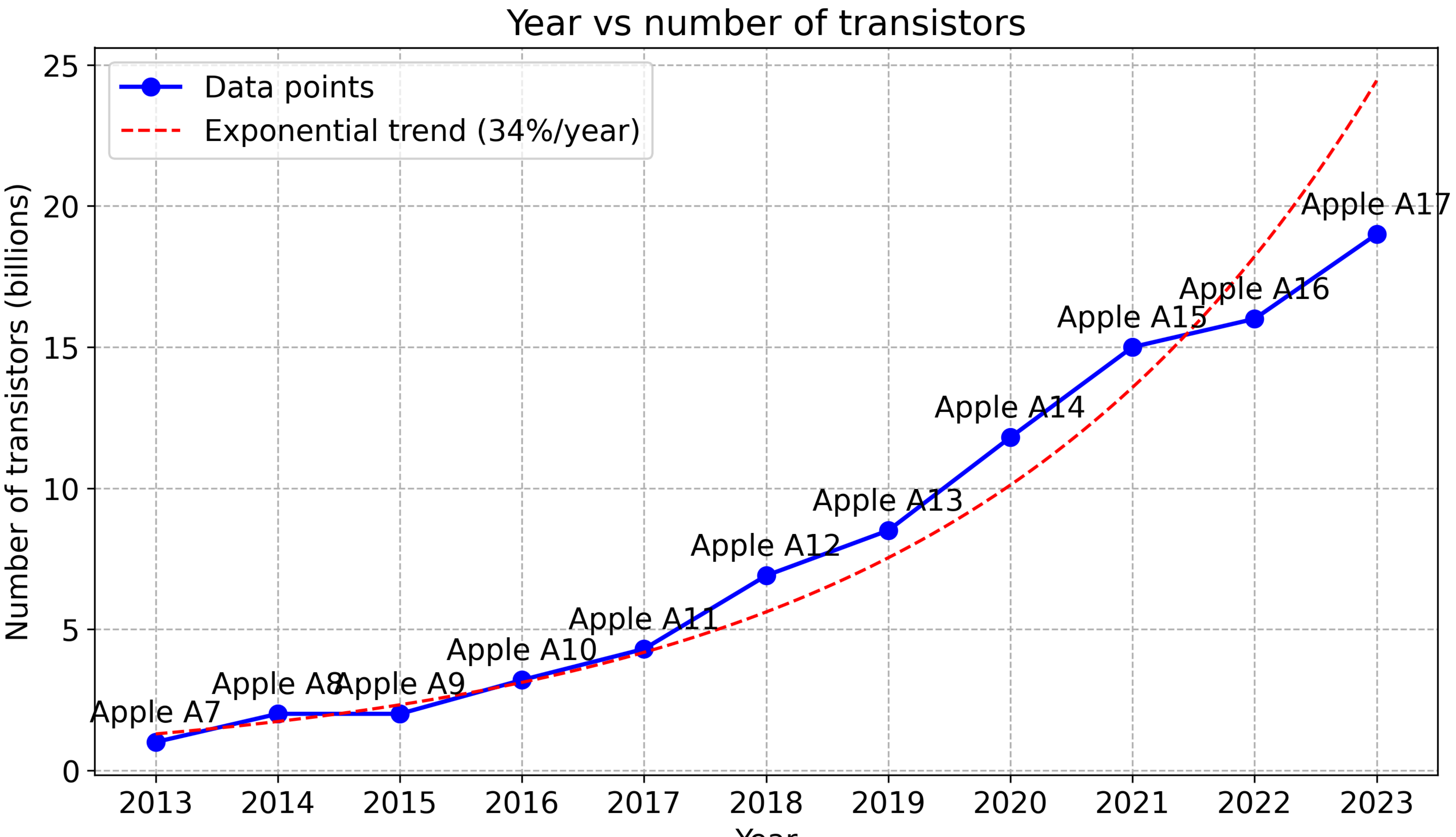
Some numbers

- Time is discrete: clock cycle
- Processors: 4 GHz (4×10^9 cycles per second)
- One cycle is 0.25 nanoseconds
- light: 7.5 centimeters per cycle
- One byte per cycle: 4 GB/s

Easily CPU bound

Frequencies and transistors

processor	year	frequency	transistors
Pentium 4	2000	3.8 GHz	0.040 billions
Intel Haswell	2013	4.4 GHz	1.4 billions
Apple M1	2020	3.2 GHz	16 billions
Apple M2	2022	3.49 GHz	20 billions
Apple M3	2024	4.05 GHz	25 billions
Apple M4	2024	4.5 GHz	28 billions
AMD Zen 5	2024	5.7 GHz	50 billions


Year vs number of transistors
Data points
Exponential trend (34%/year)
Number of transistors (billions)
Year
0
5
10
15
20
25
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
Apple A7
Apple A8
Apple A9
Apple A10
Apple A11
Apple A12
Apple A13
Apple A14
Apple A15
Apple A16
Apple A17

Where do the transistors go?

- More cores
- More superscalar execution
- Better speculative execution
- More cache, more memory-level parallelism
- Better data-level parallelism (SIMD)

Where do the transistors go?

- More cores
- More superscalar execution (more instructions per cycle)
- Better speculative execution ($\rightarrow$ more instructions per cycle)
- More cache, more memory-level parallelism ($\rightarrow$ more instructions per cycle)
- Better data-level parallelism (SIMD) ($\rightarrow$ fewer instructions)

Superscalar execution

processor	year	arithmetic logic units	SIMD units
Pentium 4	2000	2	2×128
AMD Zen 2	2019	4	2×256
Apple M*	2019	6+	4×128
Intel Lion Cove	2024	6	4×256
AMD Zen 5	2024	6	4×512

Moving to up to 4 load/store per cycle

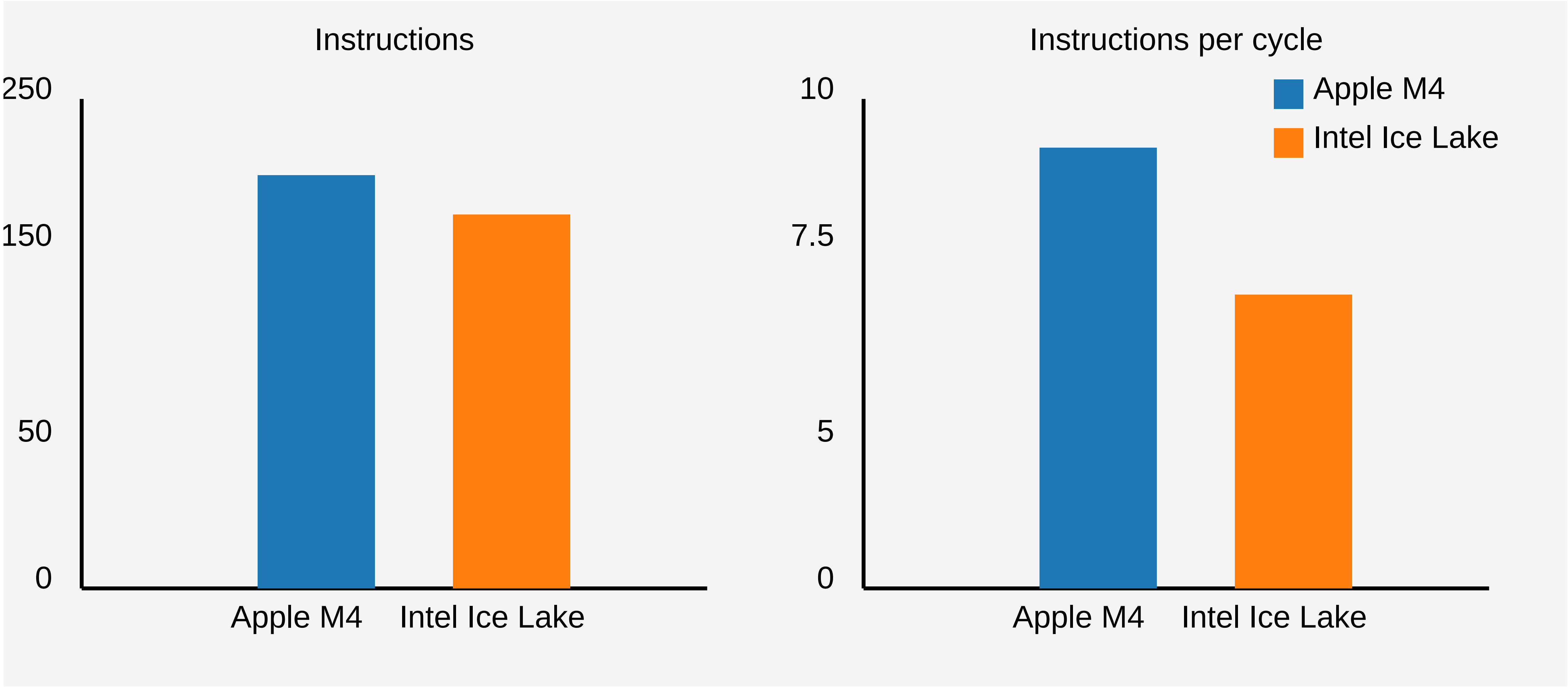
Parsing a number

- `1.3321321e-12` to `double`

```
double result;
fast_float::from_chars(
  input.data(), input.data() + input.size(), result);
```

Reference: Number Parsing at a Gigabyte per Second, Software: Practice and Experience 51 (8), 2021

Parsing a number



Lemire's Rule 1

> Modern processors execute *nearly* as many instructions per cycle as you can supply.

- with caveats: branching, memory, and input/output

Lemire's Corollary 1

> In computational workloads (batches), minimizing instruction count is critical for achieving optimal performance.

Lemire's Tips

1. Batch your work in larger units to save instructions.
2. Simplify the processing down to as few instructions as possible.

Going back to number parsing

- Our number parser: major browsers (Safari, Chrome), GCC (12+), C#, Rust
- About $4\times$ faster than the conventional alternatives.
- How did we do it?

We massively reduced the number of CPU instructions required.

function	instructions
strtod	> 1000
our parser	≈ 200

Reference:

Number Parsing at a Gigabyte per Second, Software: Practice and Experience 51 (8), 2021

SWAR

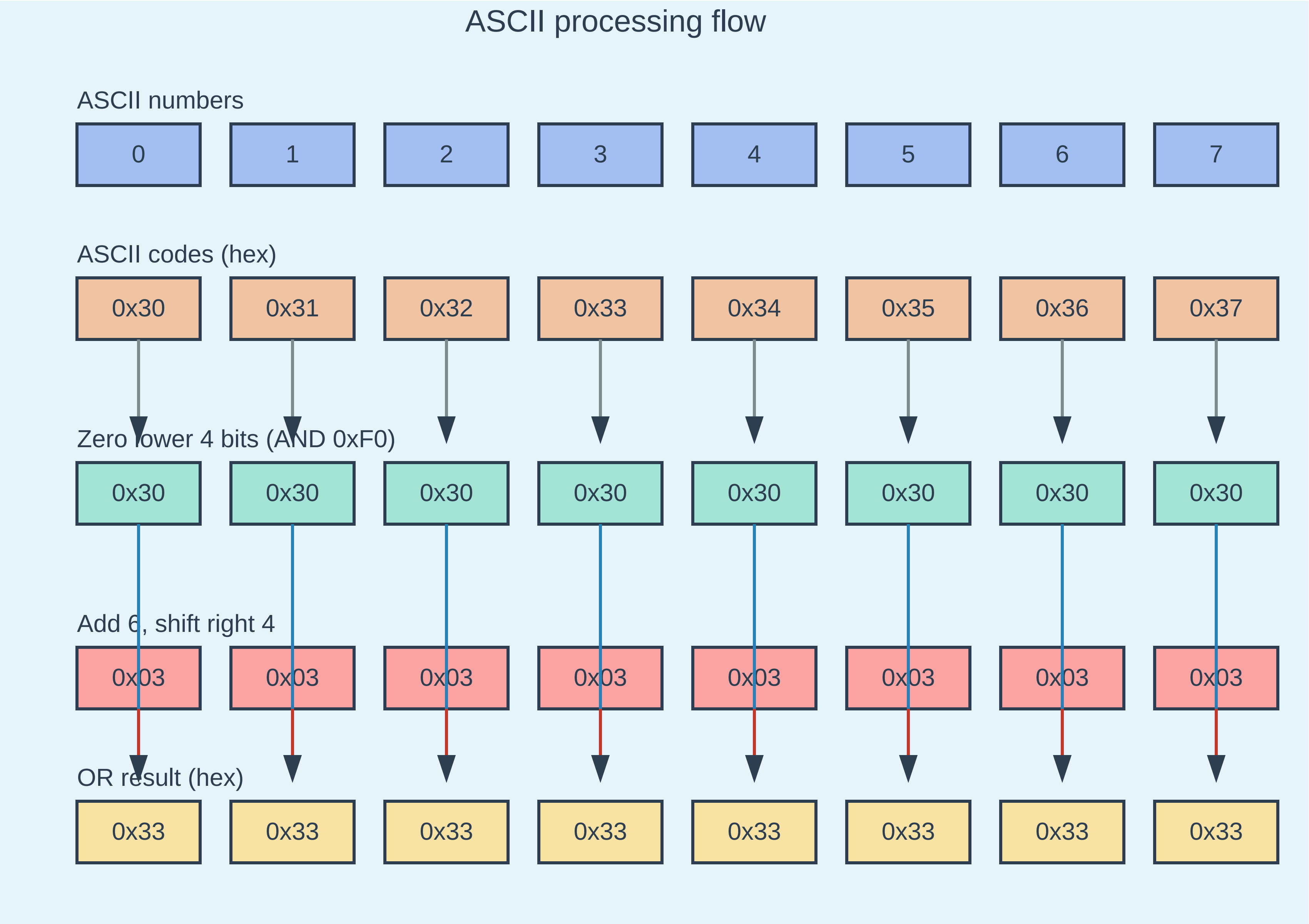
- Stands for SIMD within a register
- Use normal instructions, portable (in C, C++,...)
- A 64-bit register can be viewed as 8 bytes
- Requires some cleverness

Check whether we have a digit

In ASCII/UTF-8, the digits 0, 1, ..., 9 have values
0x30, 0x31, ..., 0x39.

To recognize a digit:

- The high nibble should be 3.
- The high nibble should remain 3 if we add 6 (0x39 + 0x6 is 0x3f)


ASCII processing flow
ASCII numbers
0
1
2
3
4
5
6
7
ASCII codes (hex)
0x30
0x31
0x32
0x33
0x34
0x35
0x36
0x37
Zero lower 4 bits (AND 0xF0)
0x30
0x30
0x30
0x30
0x30
0x30
0x30
0x30
Add 6, shift right 4
0x03
0x03
0x03
0x03
0x03
0x03
0x03
0x03
OR result (hex)
0x33
0x33
0x33
0x33
0x33
0x33
0x33
0x33

Batching (unrolling)

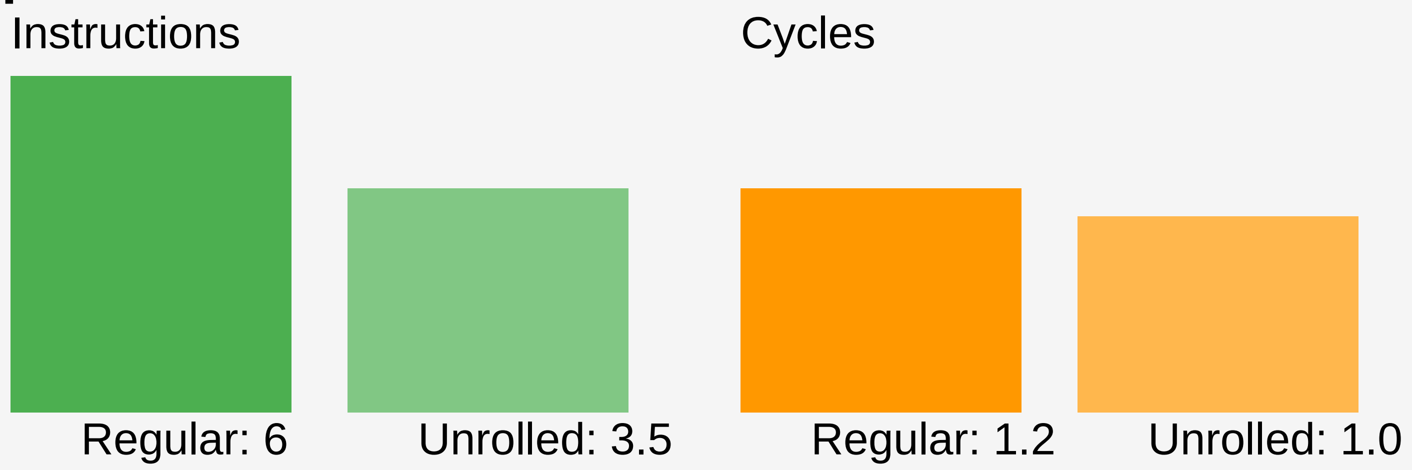
6 to 7 instructions per multiplication

```
  for (size_t i = 0; i < length; i++)
    sum +=  x[i] * y[i];
```

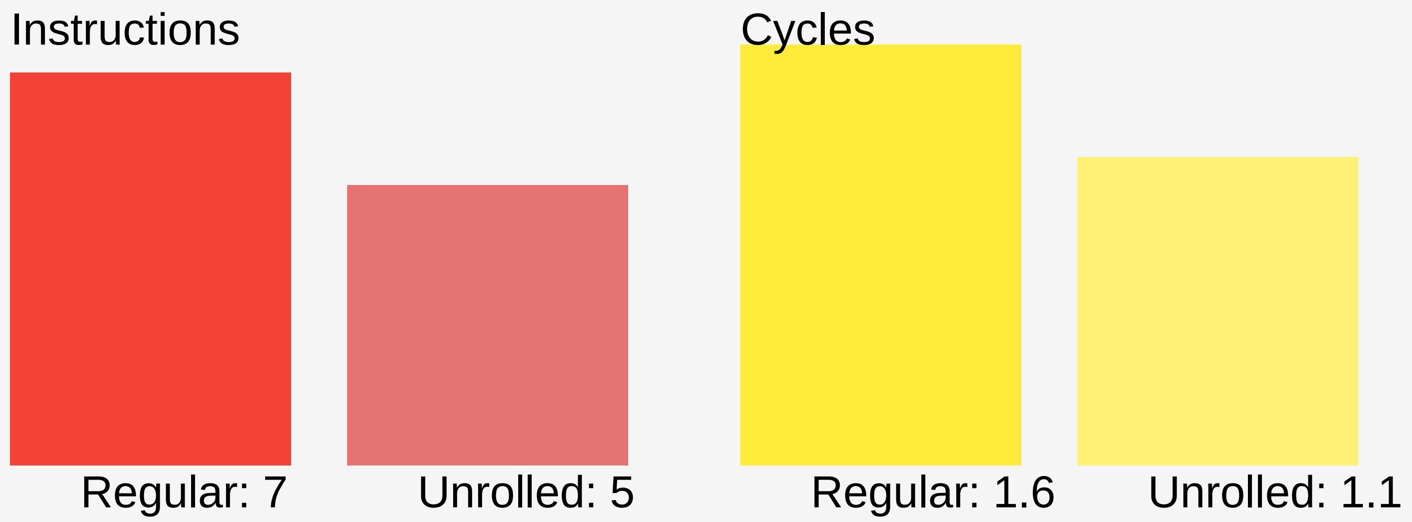
3 to 5 instructions per multiplication

```
    for (; i < length - 3; i += 4)
      sum +=  x[i] * y[i]
            + x[i + 1] * y[i + 1]
            + x[i + 2] * y[i + 2]
            + x[i + 3] * y[i + 3];
```

Apple M4



Intel Ice Lake



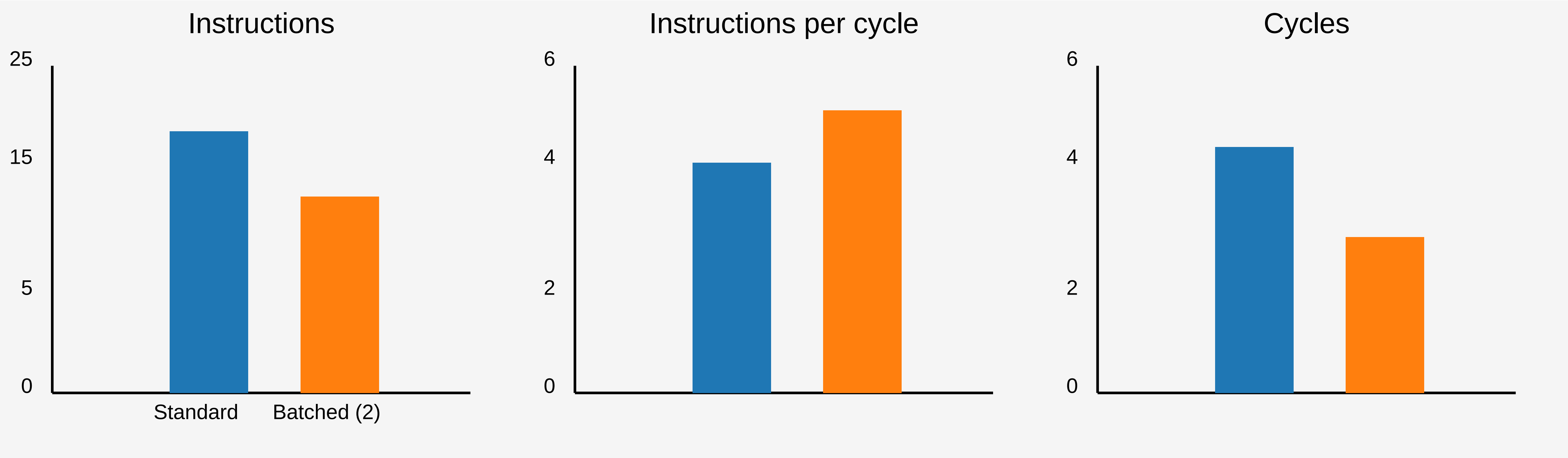
Knuth's random shuffle

```
PROCEDURE shuffle(array)
    FOR j FROM |array| - 1 DOWN TO 1
        k ← random_integer(0, j)
        SWAP array[j] WITH array[k]
    END FOR
END PROCEDURE
```

Batched random shuffle

- Draw one random number
- Compute two indices (with high probability)
- Reduces the instruction count
- Reduces the number of branches

Results (Apple M4): Use a large array (8 MB).



Reference: Batched Ranged Random Integer Generation, Software: Practice and Experience 55 (1), 2025

Branching

Hard-to-predict branches can derail performance

Unicode (UTF-16)

- Code points from U+0000 to U+FFFF, a single 16-bit value.
- Beyond: a surrogate pair `[U+D800 to U+DBFF]` followed by `U+DC00 to U+DFFF`

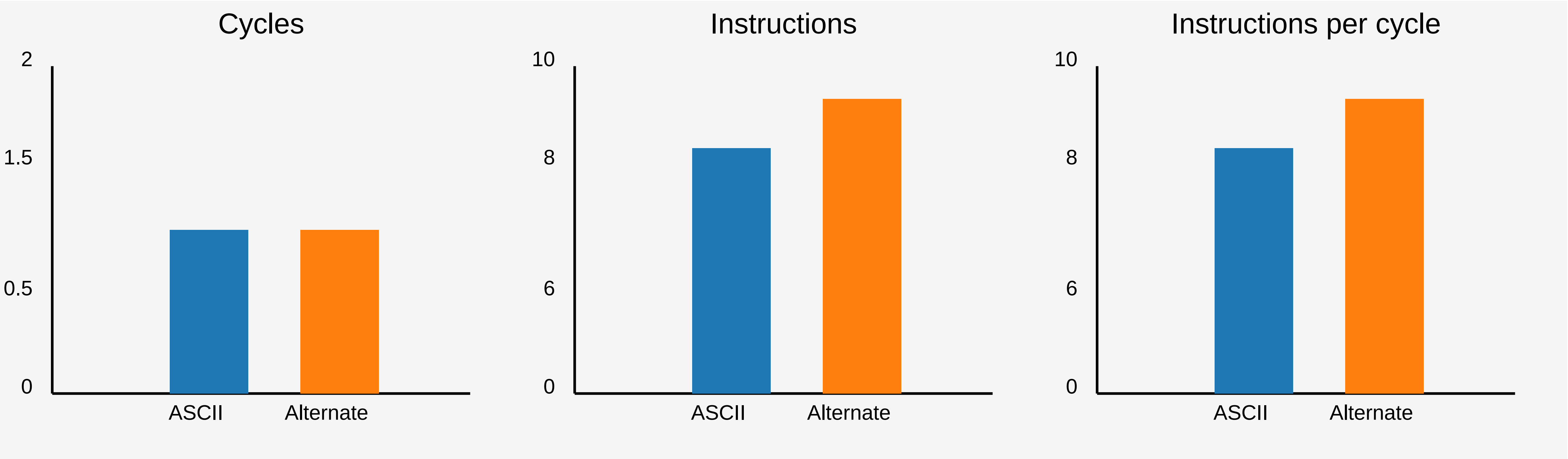
Validate

- Check for a lone code unit ($x \leq \mathrm{0xD7FF} \lor x \geq \mathrm{0xDBFF}$), if so ok
- Check for the first part of the surrogate ($\mathrm{0xD800} \leq x \leq \mathrm{0xDBFF}$) and if so check that we have the second part of a surrogate

Validate

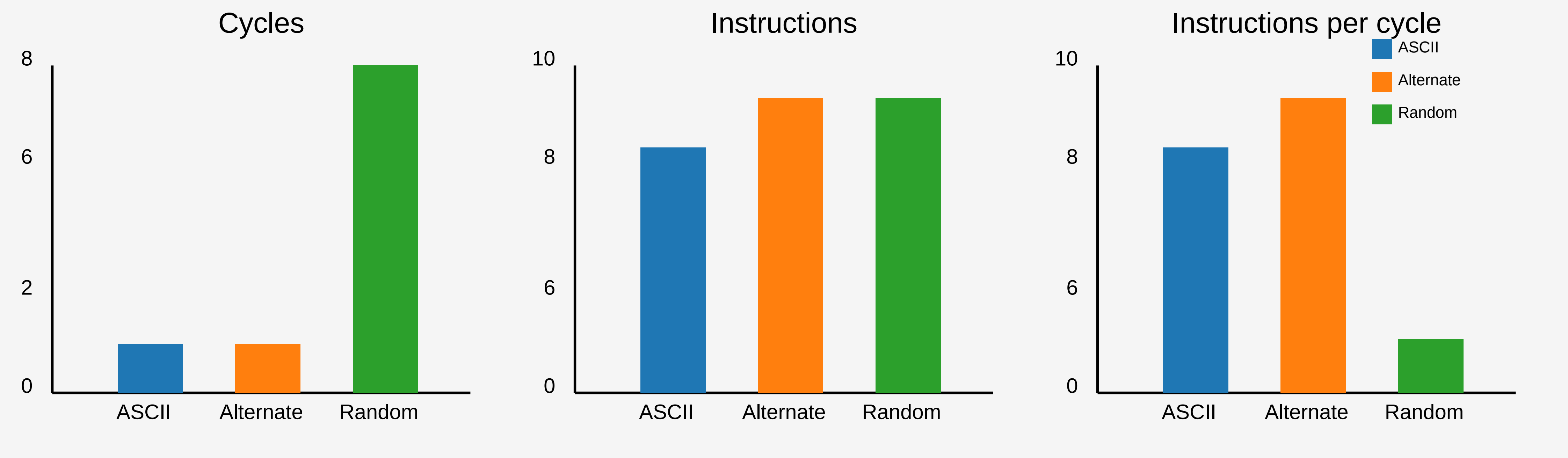
```
PROCEDURE validate_utf16(code_units)
    i ← 0
    WHILE i < |code_units|
        unit ← code_units[i]
        IF unit ≤ 0xD7FF OR unit ≥ 0xE000 THEN
            INCREMENT i
            CONTINUE
        IF unit ≥ 0xD800 AND unit ≤ 0xDBFF THEN
            IF i + 1 ≥ |code_units| THEN
                RETURN false
            next_unit ← code_units[i + 1]
            IF next_unit < 0xDC00 OR next_unit > 0xDFFF THEN
                RETURN false
            i ← i + 2  // Valid surrogate pair
            CONTINUE
        RETURN false
    RETURN true
```

Performance results (Apple M4)



1 character per cycle might be just 4 GB/s (slower than disk)

Performance results (Apple M4)

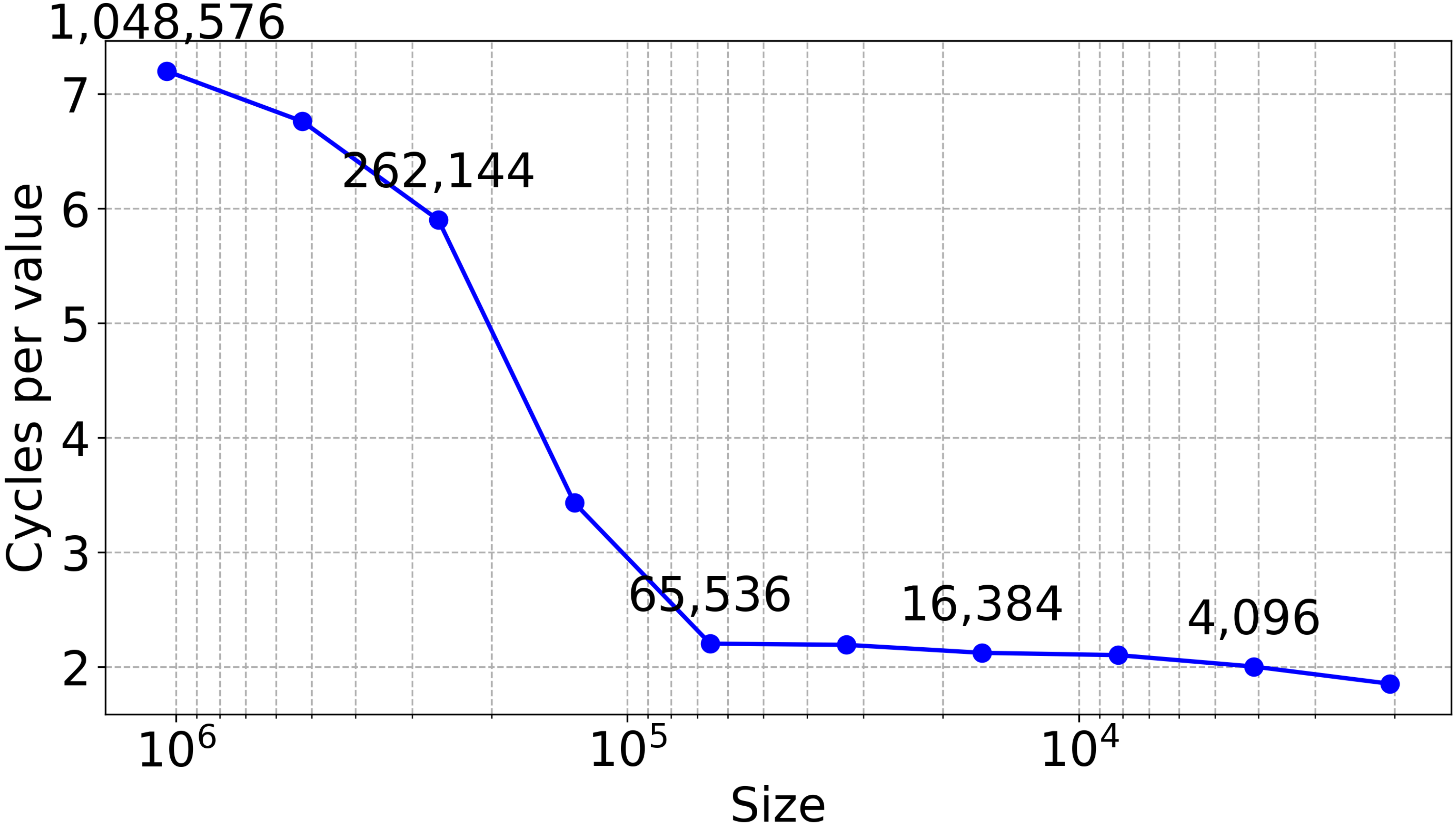


We are now barely at 1 GB/s!

Speculative execution

- Processors *predict* branches
- They execute code *speculatively* (can be wrong!)

How much can your processor learn?



1,048,576
262,144
65,536
16,384
4,096
Cycles per value
7
6
5
4
3
2
10^6
10^5
10^4
Size

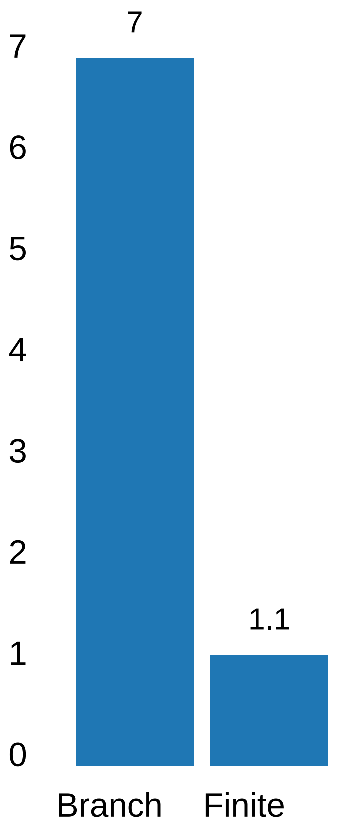
Finite state machine to the rescue

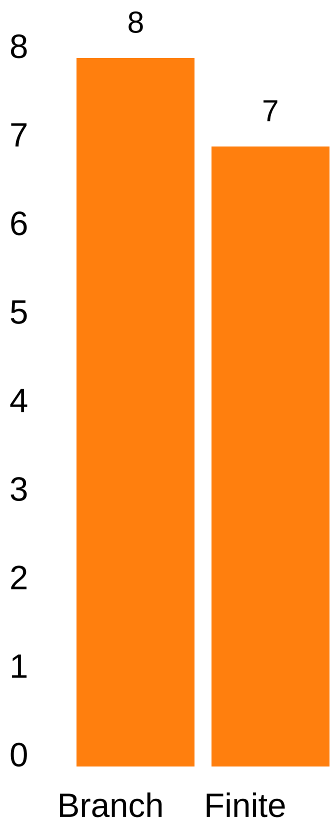
- Can identify characters by the most significant 8 bits.
- Trivial finite state machine: default, has just encountered a high surrogate, or error.

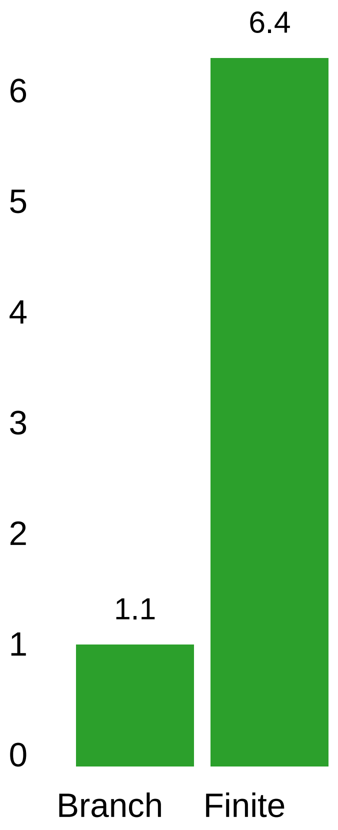
```cpp
static uint8_t transition_table[3][256] = {
    {...},
    {...},
    {...}
};

bool is_valid_utf16_ff(std::span<uint16_t> code_units) {
    uint8_t state = 0; // Start in Initial state
    for (auto code_unit : code_units) {
        uint8_t high_byte = code_unit >> 8;
        state = transition_table[state][high_byte];
    }
    return state == 0; // Valid only if we end in Initial state
}
```

Performance results (Apple M4)

The finite-state approach can be $7\times$ faster!



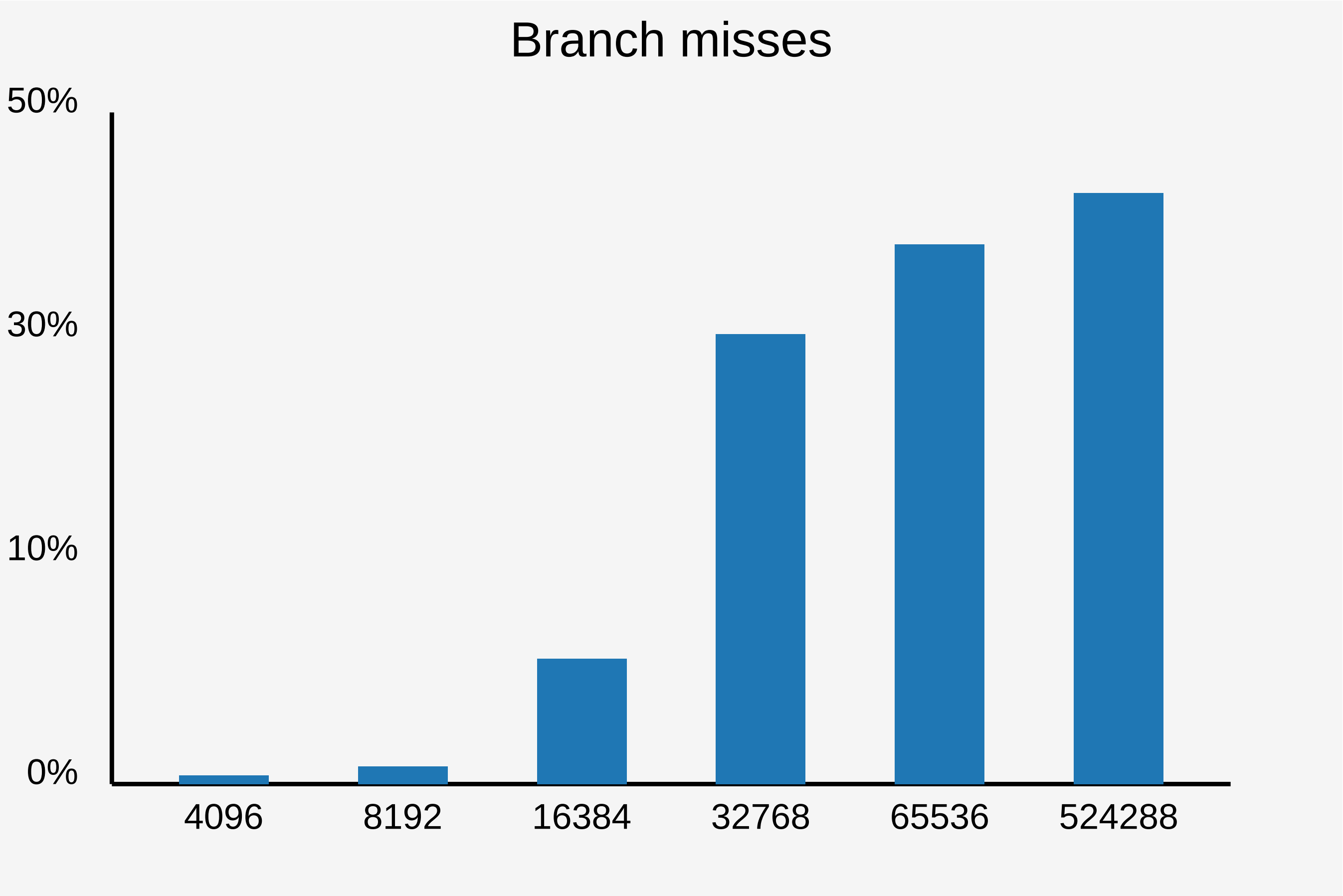




Rules of thumb

1. Processors can 'learn' thousands of branches: benchmark over massive inputs.
2. Pick a solution without branches when it provides the same performance.

Apple M4 can learn 10,000 random (0/1) branches.



Pipelining

How does the processor manage to validate one UTF-16 character per cycle when it takes **many cycles** just to *load* the character?

cycle	action	action	pizza en route
1	order pizza A		
2	order pizza B		A🚚
3	order pizza C		A🚚, B🚚
4	order pizza D	eat pizza A 🍕	B🚚, C🚚
5	order pizza E	eat pizza B 🍕	C🚚, D🚚
6	order pizza F	eat pizza C 🍕	D🚚, E🚚

Little's Law

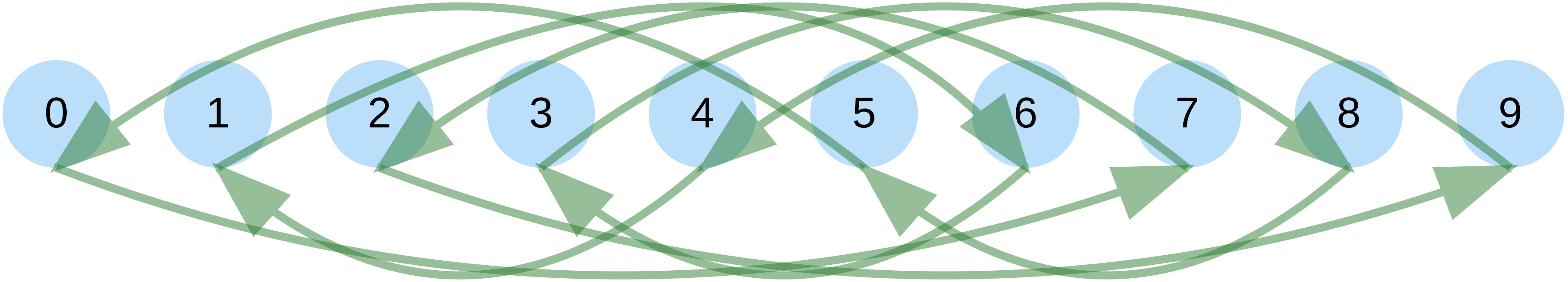
- Latency harms throughput
- Parallelism hides latency

$$\text{throughput} = \frac{\text{parallelism}}{\text{latency}}$$

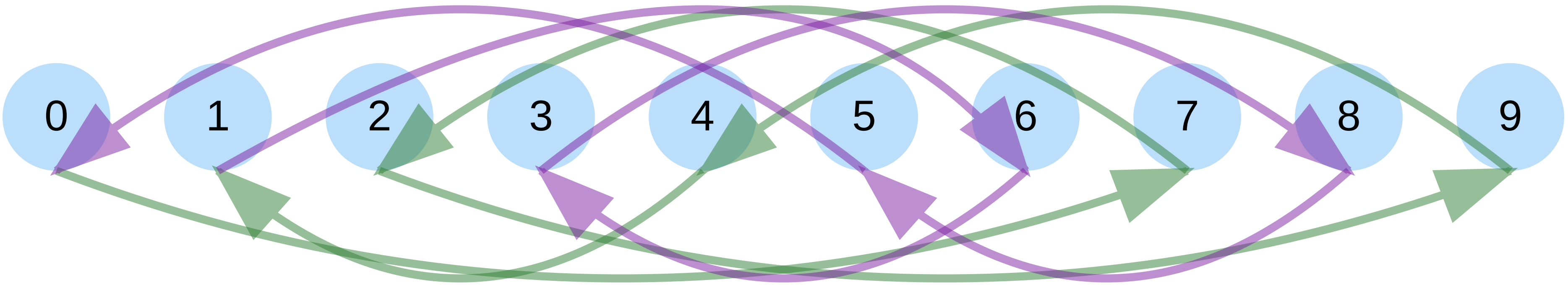
Memory-level parallelism

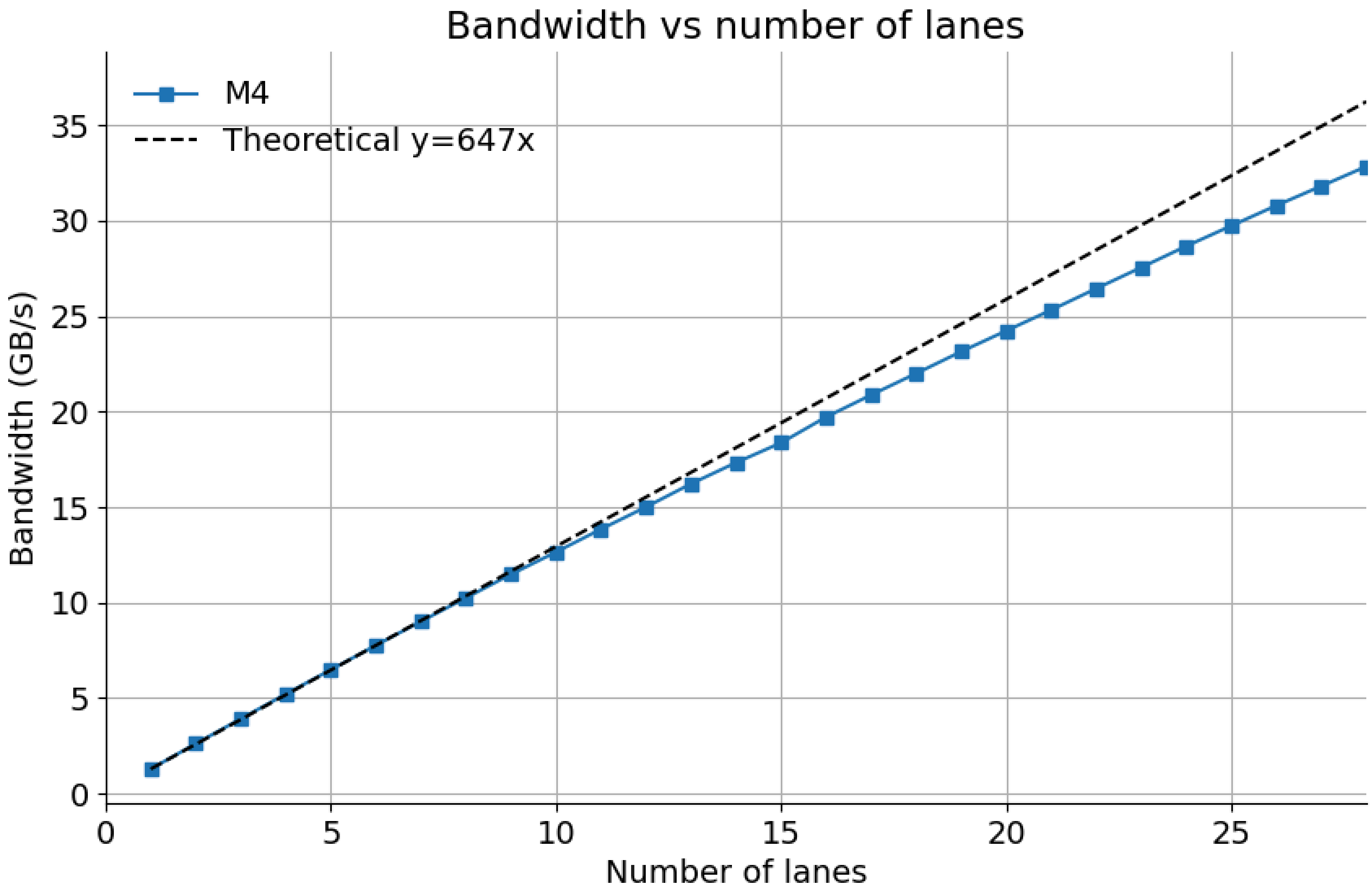
- Create large array of indices forming a cycle
- Start with [0,1,2,3,4]
- Shuffle so that no index can remain in place.
- Start with [4,0,3,2,1]
- Sandra Sattolo's algorithm
- This forms a random path

1 lane



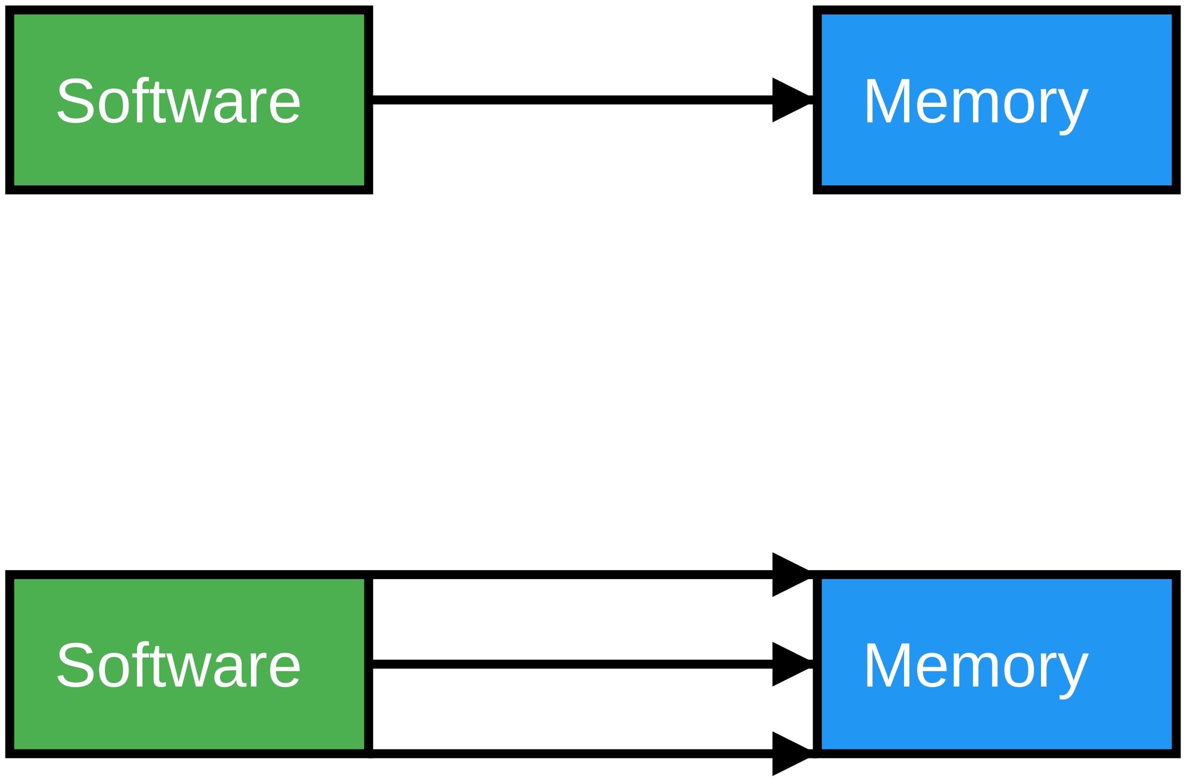
2 lanes



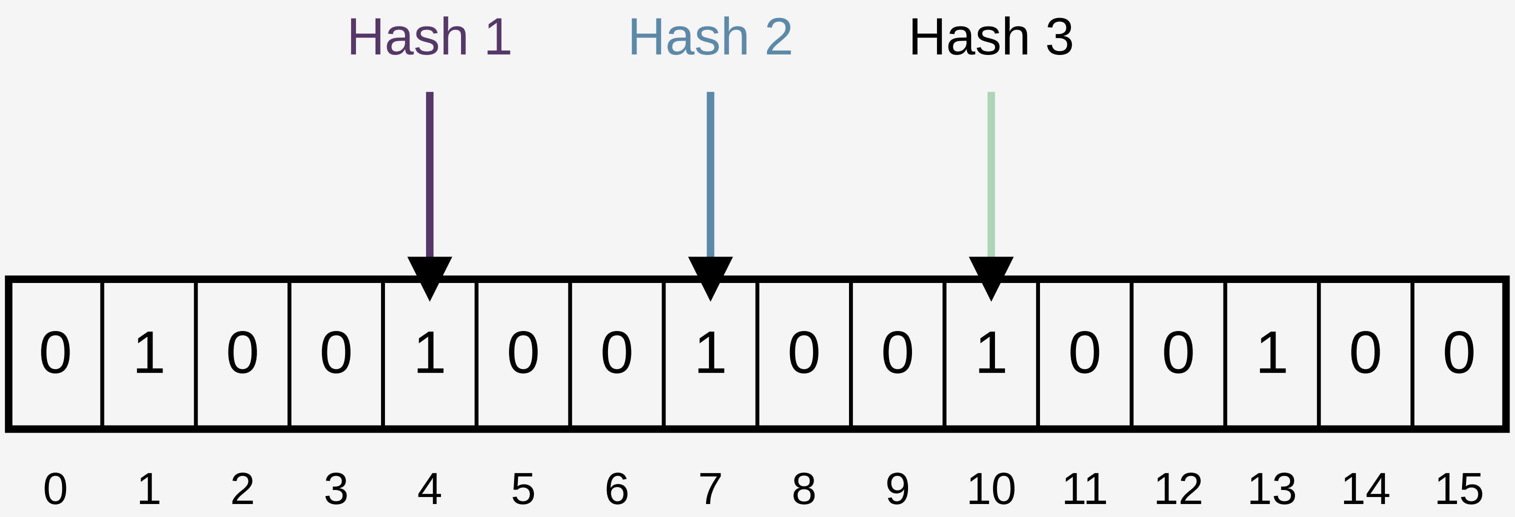


Bandwidth vs number of lanes
M4
Theoretical y=647x
Bandwidth (GB/s)
Number of lanes
0
5
10
15
20
25
30
35
0
5
10
15
20
25

Consequence

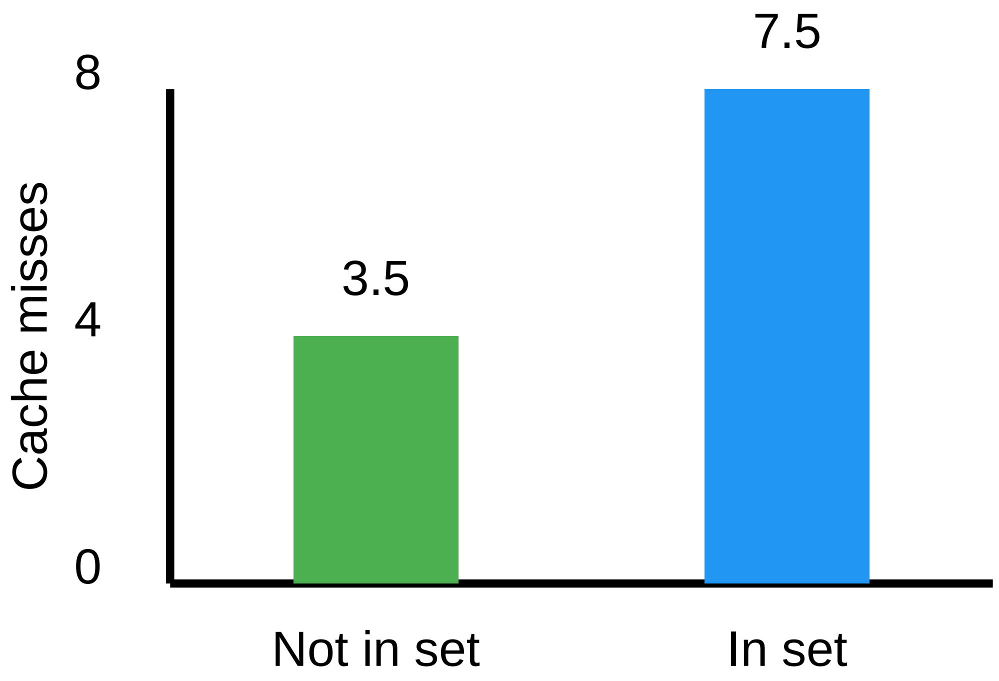


Bloom filter



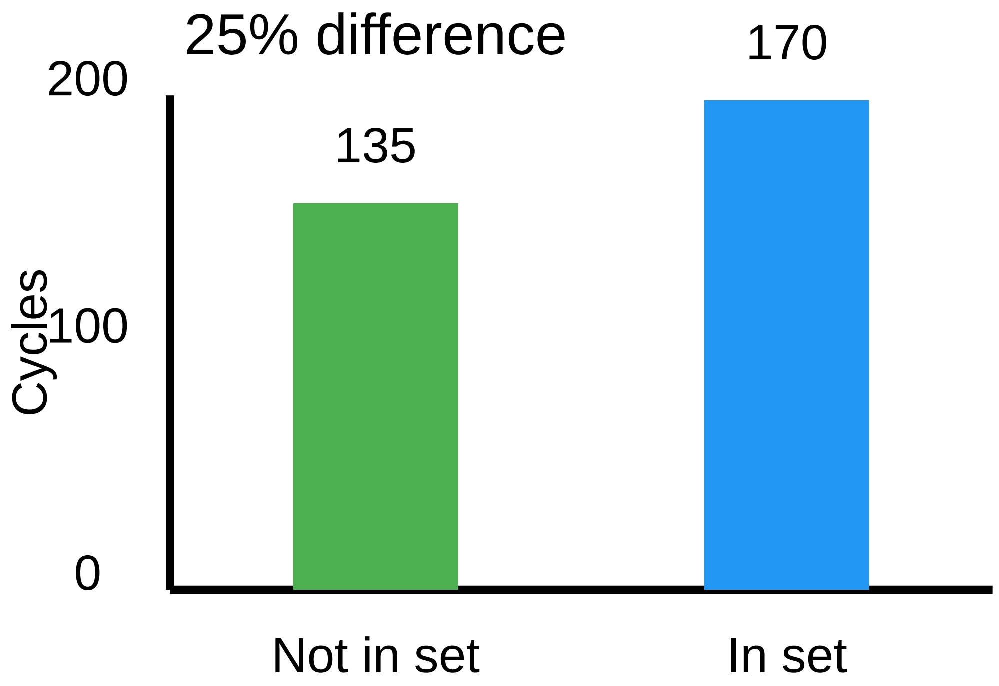
Bloom filter

8 hash functions, (Intel Ice Lake processor, out-of-cache filter)



Less than half the cache misses

Bloom filter



Data-level parallelism

SIMD

- Stands for Single instruction, multiple data
- Allows us to process 16 bytes or more with one instruction
- Supported on all modern CPUs (phone, laptop)

ASCII to lower case

```
For each character c
    If c - 'A' < 'Z' - 'A' then
        c = c + 'a' - 'A'
    EndIf
EndFor
```

ASCII to lower case: 64 characters in 3 instructions

- Compute $c - \mathrm{A}$

```
__m512i ca = _mm512_sub_epi8(c, _mm512_set1_epi8('A'));
```

- Turn $c - '\mathrm{A}' \leq \mathrm{Z} - \mathrm{A}$ into a mask

```
__mmask64 is_upper = _mm512_cmple_epu8_mask(ca, _mm512_set1_epi8('Z' - 'A'));
```

- Add $\mathrm{a} - \mathrm{A}$ to c according to mask

```
__m512i to_lower = _mm512_mask_add_epi8(c, is_upper, c, to_lower)
```

Deltas (C)

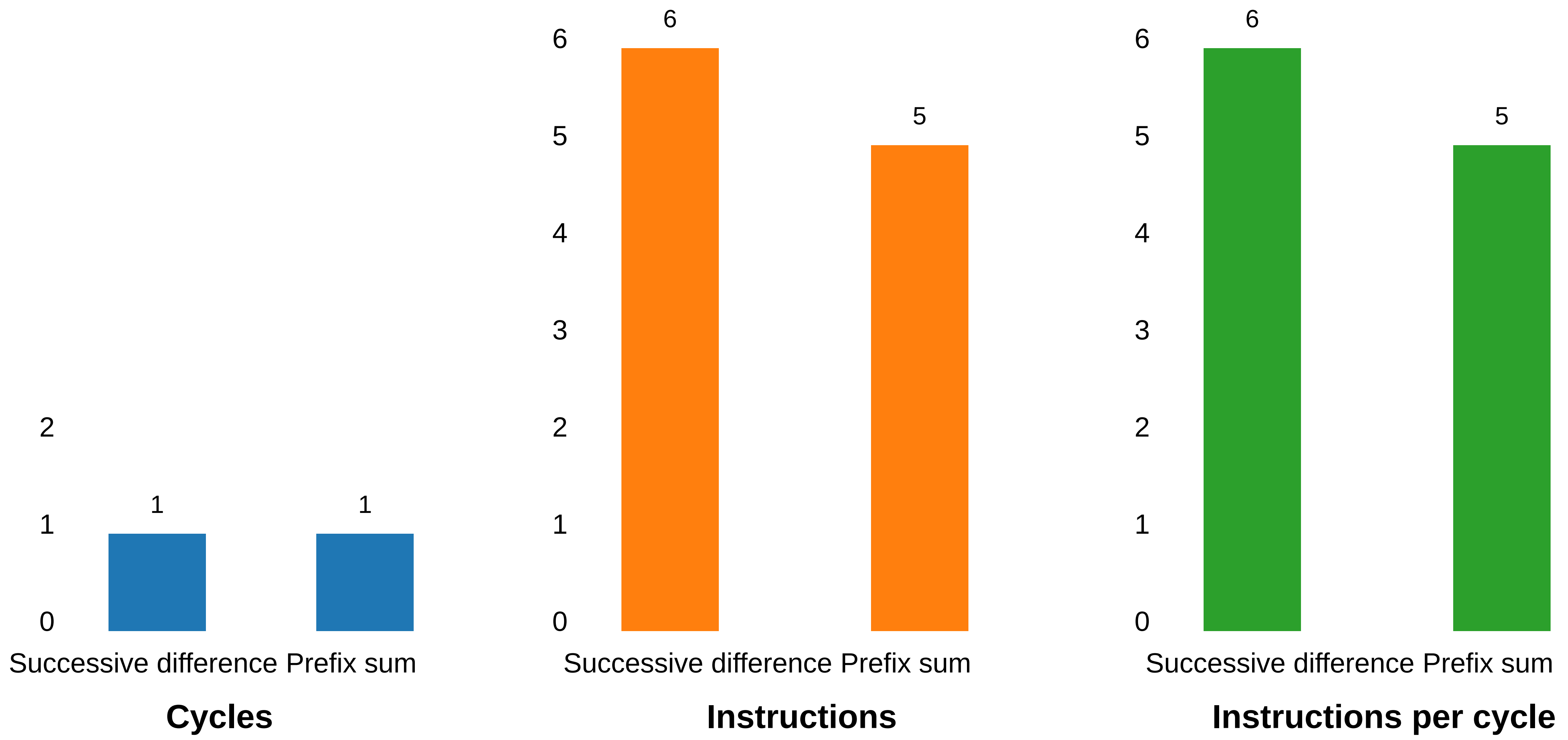
successive difference:

```
    for (size_t i = 1; i < n; ++i) {
        dst[i] = src[i] - src[i - 1];
    }
```

prefix sum:

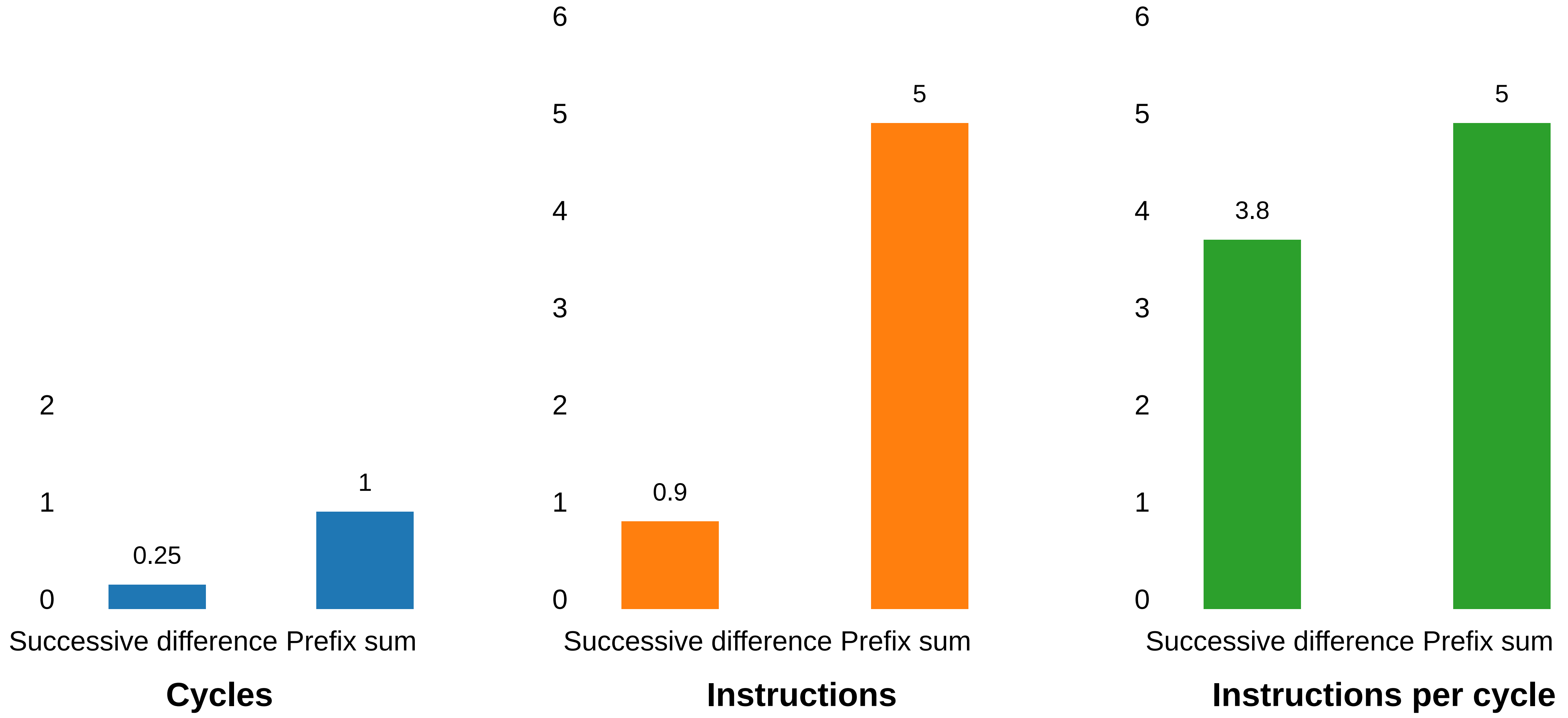
```
    for (size_t i = 1; i < n; ++i) {
        dst[i] = dst[i - 1] + src[i];
    }
```

Apple M4



2
1
0
1
1
Successive difference
Prefix sum
Cycles
6
5
4
3
2
1
0
6
5
Successive difference
Prefix sum
Instructions
6
5
4
3
2
1
0
6
5
Successive difference
Prefix sum
Instructions per cycle

Now allow SIMD! (Autovectorization)

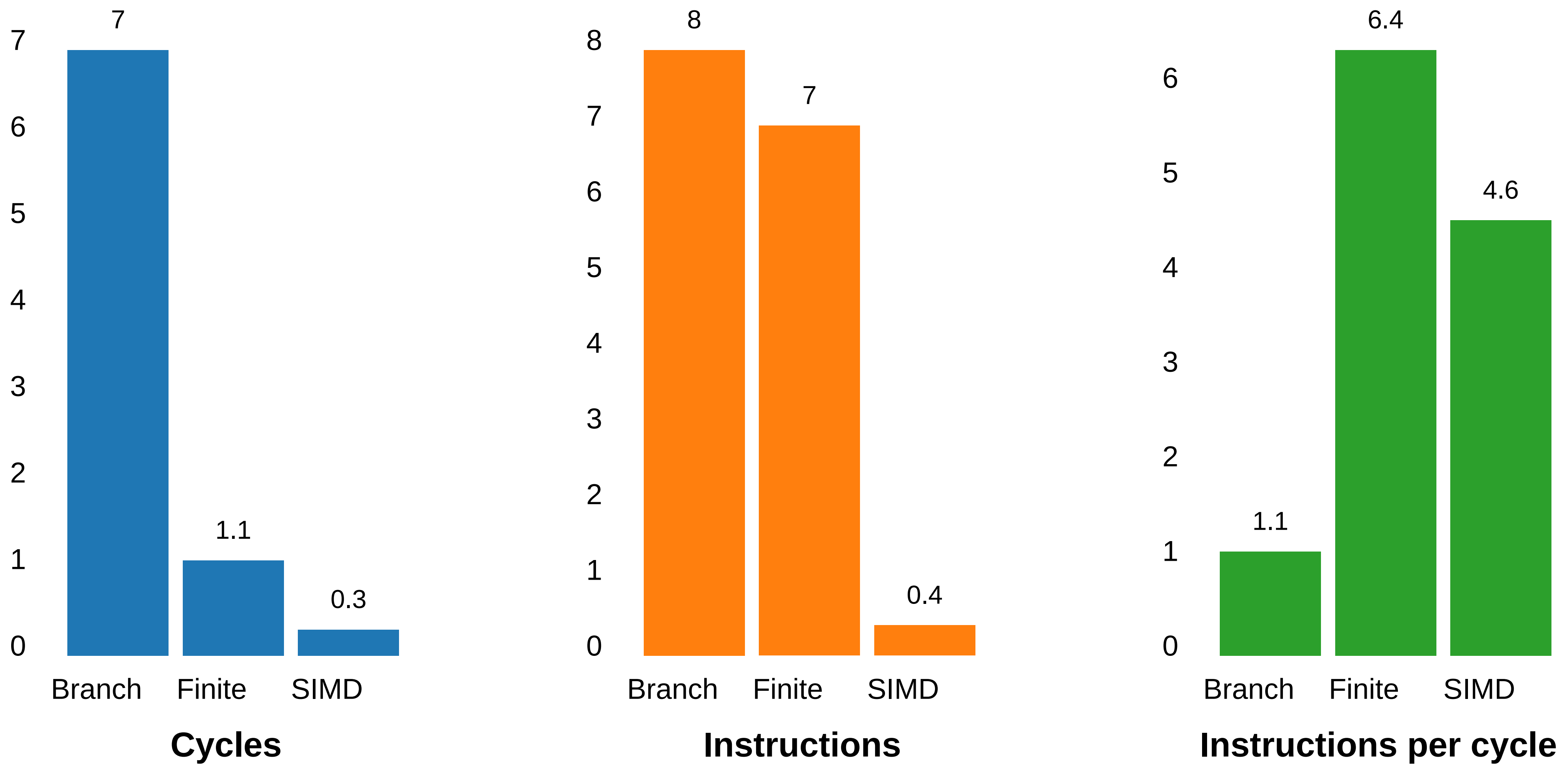


Need to learn SIMD design magic !

UTF-16

- Write SIMD correction function (not just validation)
- Actually deployed in v8 (Google Chrome, Microsoft Edge)

UTF-16, random (adversarial), Apple M4



- Most x64 processors have AVX2 (32-byte register)
- AMD Zen 5 has powerful AVX-512 (64-byte register)
- ARM has NEON + SVE/SVE2
- RISC-V has its vector instructions
- Loonson processes have AVX2-like instructions

Interested? Check these projects

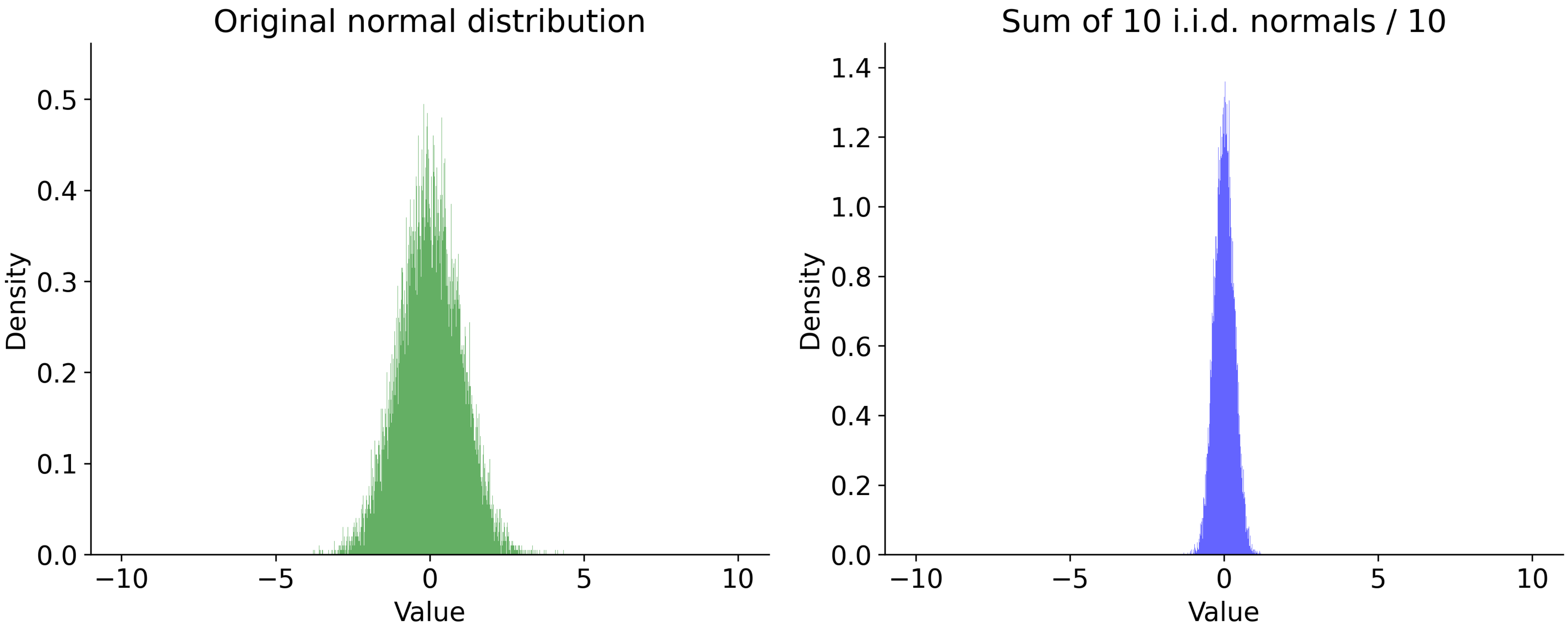
- simdjson: The fastest JSON parser in the world https://simdjson.org
- simdutf: Unicode routines (UTF8, UTF16, UTF32) and Base64 https://github.com/simdutf/simdutf

Measurements

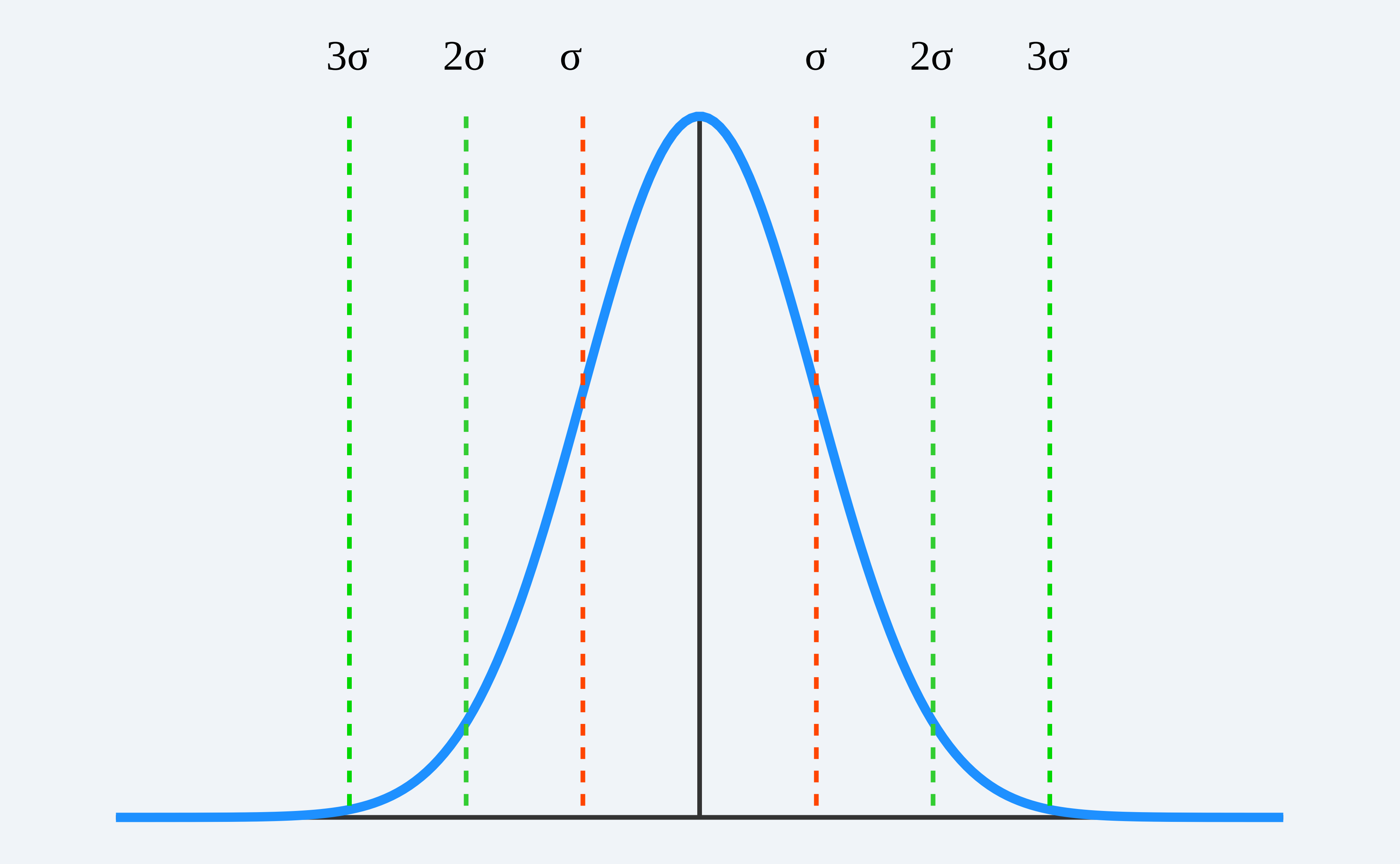
- We often assume that measurements (timings) are normally distributed.
- It is often an incorrect assumption.

Measurements

- If your measurements are normally distributed, the 'error' falls off as $1/\sqrt{N}$


Original normal distribution
Density
Value
0.0
0.1
0.2
0.3
0.4
0.5
−10
−5
0
5
10
Sum of 10 i.i.d. normals / 10
Density
Value
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
−10
−5
0
5
10

Sigma events



- 1-sigma is 32%

- 1-sigma is 32%
- 2-sigma is 5%

- 1-sigma is 32%
- 2-sigma is 5%
- 3-sigma is 0.3% (once every 300 trials)

- 1-sigma is 32%
- 2-sigma is 5%
- 3-sigma is 0.3% (once every 300 trials)
- 4-sigma is 0.00669% (once every 15000 trials)

- 1-sigma is 32%
- 2-sigma is 5%
- 3-sigma is 0.3% (once every 300 trials)
- 4-sigma is 0.00669% (once every 15000 trials)
- 5-sigma is 5.9e-05% (once every 1,700,000 trials)
- 6-sigma is 2e-07% (once every 500,000,000)
- $e^{-n^2/2}/(n * \sqrt{\pi/2}) \times 100$ for $n > 3$

What if we dealt with log-normal distributions?

Real-world measurements

- You cannot assume normality
- Measurements are **not independent**.
- Reality: the absolute minimum is often a *reliable metric*
- Margin: difference between mean and minimum

Conclusion

- Processors are getting much better! Wider!
- 'hot spot' engineering can fail, better to reduce overall instruction count.
- Branchy code can do well in synthetic benchmarks, but be careful.